

A drone that flies itself through a building — and hands you the 3D model.

AetherScan autonomously explores unknown indoor spaces with no GPS, maps them in real time, and exports a measured 3D reconstruction. The autonomy — perception, SLAM, exploration and flight control — is built from scratch and validated end to end.



Live capture — autonomous scan of a real indoor space (Meta Replica 'apartment_1'), single ~5-minute run.

87.8%

area covered

353K

points captured

58.8 m²

floor mapped

0

GPS satellites

WHAT WORKS TODAY

- Fully autonomous, GPS-denied**
Frontier + coverage exploration over a log-odds occupancy grid the drone builds itself — it decides where to fly to finish the map.
- Flies on its own estimate**
On-board SLAM (correlative scan-matching) corrects pose drift; flying on the estimated pose reaches coverage parity with ground truth.
- Real-time reconstruction + export**
Dense semantic point cloud, exportable as coloured PLY, Poisson GLB mesh and an SVG floor plan with dimensions.
- Validated flight stack**
6-DoF rigid-body physics at 500 Hz, cascaded controller: ~10-14° peak tilt, ±8 mm altitude hold across scenes.

Honest status

Everything above is real output from the system running now. It is validated in a high-fidelity simulator with realistic sensor noise and pose drift. The aircraft is fully designed (below) but not yet physically built — that is what we are raising for.

HOW IT WORKS



THE AIRCRAFT



CLASS 360 mm quad-X · 7" guarded
MASS / ENDURANCE 1.45 kg · ~14-17 min
SENSORS RPLIDAR A2 · D435i · flow/ToF · IMU
COMPUTE Jetson Orin Nano + Pixhawk 6C / PX4
BOM ~\$1,659 · CAD + PCB defined in code

WHY IT WINS

The moat is GPS-denied indoor autonomy. Most drones need GPS or a pilot; AetherScan needs neither.

- vs Skydio / DJI outdoor, GPS-reliant
- vs Flyability caged, pilot-flown
- vs Matterport / NavVis tripod, manual

FOCUS MARKET

Construction-progress capture & facility survey — daily, repeatable, GPS-denied interior 3D where speed and autonomy beat tripods.

PROVEN IN SIMULATION TODAY



ROADMAP TO FIRST FLIGHT



THE ASK

Seed capital funds the first physical build. Bench-first staging reaches a fully autonomous indoor scan for ~\$1,060 of parts. Still ahead and scoped: real-airframe bring-up, global loop closure, EMC / thermal bench tests.

github.com/sypherdark/AetherScan

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